

TSCR SERIES

MODEL

TSCR-05E

The TSCR-05E collaborative robot brings automation to workspaces shared with people. Harmonic-drive joints, hand-guided teaching and safety-rated monitored stop let it run fence-free machine tending, assembly, testing and packaging cells. A compact 1052 mm build height and 138 × 138 mm base make it easy to place beside operators on existing benches.

6-Axis	Repeatability	Texsonics RC	Site-ready
Payload 5 kg · Reach 800 mm	±0.03 mm positional accuracy	In-house controller, drives and teach pendant	Floor / Wall / Ceiling mount, factory calibrated

01 SPECIFICATIONS

MECHANICAL

AXES	6
PAYLOAD	5 kg
MAX REACH	800 mm
REPEATABILITY	±0.03 mm
JOINTS	Harmonic drive, force-limited

ENVIRONMENT / CONTROL

HEIGHT	1052 mm
BASE SIZE	138 × 138 mm
TEACHING	Hand guiding + teach pendant
SAFETY	Collaborative operation per ISO/TS 15066 design intent
PROGRAMMING	Graphical HMI / Offline CAM

02 PRODUCT



TSCR-05E · TSCR SERIES

03

WORKING RANGE

MOTION — GROUP A

J1 — BASE ROTATION	±170°
J2 — SHOULDER	±170°
J3 — ELBOW	±160°

MOTION — GROUP B

J4 — WRIST 1	±170°
J5 — WRIST 2	±170°
J6 — TOOL ROTATION	±175°

04

OVERVIEW

WORKS WITH PEOPLE,
NOT BEHIND FENCES.

The TSCR-05E is a collaborative robot with force-limited harmonic-drive joints, hand-guided teaching and safety-rated monitored stop. Deployable on existing benches in days, not months.

01

HAND-GUIDED TEACHING

Move the arm through the motion and record waypoints — no programming language required.

02

FORCE-LIMITED JOINTS

Every joint monitors torque and halts on unexpected contact, per the design intent of ISO/TS 15066.

03

SMALL FOOTPRINT

A 800 mm reach in a 138 × 138 mm base fits beside operators on existing lines.

05

CONTROL & INTEGRATION

CONTROLLER

Texsonics RC series
in-house motion controller,
dual-kernel real-time OS

TEACH PENDANT

11.5" touch tablet
drag-teaching, Wi-Fi,
on-device offline
programming

FIELDBUS

**EtherCAT / Modbus
TCP**
industrial fieldbus,
dual-protocol support

SOFTWARE

TEXCAM offline CAM
toolpath generation, digital
twin simulation, vision

06

APPLICATIONS

Deployable across the primary applications the TSCR-05E was engineered for — full production-line integration with the Texsonics control stack.

Fence-free Machine Tending	Assembly	Testing & Inspection	Packaging
Machine Tending	CNC Loading	Palletizing	Vision Inspection

07

BUILT FOR THE OPERATOR

- Live 3D position display
- Online command execution
- Position editor
- Variable monitor
- I/O monitor
- Error diagnostics
- Dynamic teach-in
- Digital twin simulation

08

SERVICE & SUPPORT**Made in Coimbatore.
Supported from
Coimbatore.**

Every TSCR-05E is engineered, built and calibrated in our Coimbatore facility. Field service, spares and applications engineering ship from the same address — no import lead times, no time-zone escalations.

STANDARD WARRANTY

12 months on the mechanical assembly, 12 months on the controller.

APPLICATIONS ENGINEERING

Cell design, cycle-time studies and end-of-arm tooling by our in-house team, included on every project.

SPARES & FIELD SERVICE

Direct-to-plant spares from Coimbatore, with pan-India field service engineers on 24-hour dispatch.

09

READY TO INTEGRATE**Talk to an engineer about
integrating the TSCR-05E
into your line.**

Applications engineering, cycle-time sizing, and turnkey cell design are included on every project.

CONTACT**Dharmar R**

Managing Director

+91 94426 24304

dharmar@texsonics.net

www.texsonics.net

SPECIFICATIONS AND VISUALS MAY CHANGE WITHOUT NOTICE AS PART OF ONGOING PRODUCT IMPROVEMENTS. IMAGES ARE FOR REFERENCE ONLY.
UNAUTHORIZED REPRODUCTION OF THIS CATALOGUE IS PROHIBITED WITHOUT WRITTEN PERMISSION FROM TEXSONICS SYSTEMS INDIA PRIVATE LIMITED.

TSCR-05E

Texsonics Systems India Private Limited

1/6-1, Keerakaran Thottam, Keeranatham, Coimbatore 641035 • Ph: +91 94426 24304

TSCR SERIES

6-AXIS • PAYLOAD 5 kg • REACH 800 mm

dharmar@texsonics.net • www.texsonics.net